Association of Emergency Department and Medical-Surgical Nurse Staffing on Patient Departures Against Medical Advice

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Contribution to Emergency Nursing Practice

- Patient departures against medical advice (AMA) are an unfavorable patient outcome. It is unknown whether improving nurse staffing ratios is a potential approach to reducing AMA rates in the emergency department.
- Each additional patient added to a nurse's workload in emergency departments and medical-surgical units was associated with higher rates of ED AMA departures.
- AMA departures in emergency departments are influenced not only by emergency nurse staffing levels but also by inpatient nurse staffing. AMA rates are increasing over time as emergency departments experience overcrowding. Nurse staffing is a modifiable solution to potentially reduce this patient safety risk.

Abstract

Introduction: A patient departure against medical advice is an adverse event driven by patient care dissatisfaction and is associated with higher odds of hospital readmissions and death. Given evidence that better hospital nursing resources are associated with improved patient satisfaction, we evaluated whether nurse staffing ratios are associated with rates of emergency department patient departures against medical advice.

Methods: Cross-sectional design using data from 3 sources: 2023 Penn Nurses4All survey, the 2023 Agency for Healthcare Research and Quality Healthcare Cost and Utilization Project State Emergency Department Database of patient emergency department visits in New Jersey and Oregon, and the American Hospital Association Annual Survey. Patient-to-nurse staffing ratios were measured among direct care nurses in emergency departments and medical-surgical units. Logistic regression models were constructed with nurses and patients clustered within hospitals, with adjusted models accounting for patient and hospital characteristics.

Results: The sample included 2,211,244 emergency department patient visits in 74 hospitals. After adjusting for patient and hospital characteristics, each additional patient added to an emergency department and a medical-surgical nurse's workload was associated with 19% and 55% higher odds of an against medical advice departure, respectively (emergency department, adjusted odds ratio, 1.19; 95% CI, 1.09-1.31; $P <$ .001; medical-surgical, adjusted odds ratio, 1.55; 95% CI, 1.33-1.81; $P <$ .001).

Discussion: Increases in nurses' workloads, both in the emergency department and inpatient setting, were associated with more emergency department patients leaving against medical advice. When nurses care for more patients at a time, patients are attended to less frequently, which can result in more self-directed patient departures. Insufficient staffing in medical-surgical units contributes to longer length of stay and lower inpatient bed turnover. Hospital investments in safer staffing may reduce against medical advice departures.

Key words: Emergency department; Staffing; Against medical advice; Crowding

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Introduction

Self-directed patient departures against medical advice (AMA) are considered a "never-event" as they indicate a missed opportunity to provide necessary care to patients

in need of health care services.[1] They occur when patients leave a health care setting before care providers recommend a discharge.

AMA departures can present a liability to health care systems. Patients who leave a visit AMA are 4 times as likely to experience an avoidable hospital readmission, and experience more repeat ED visits and higher morbidity and mortality rates.[2-4] A study by Tan et al[5] estimates that, in a year, inpatient hospitalizations among patients who were readmitted after an AMA departure amounted to 400,000 days and $822 million for hospitals.

Previous work suggests that AMA departures are indicative of the clinical environment where patients receive care. In particular, previous studies demonstrate that the primary reasons patients leave AMA are prolonged ED wait times and dissatisfaction with care services.[2,3] Most interventions to reduce hospital AMA rates have focused on individual patient factors: improving patient-provider communication, informing patients on reasons for delays, and calling patients after AMA departures to ensure proper clinical follow-up.[6] Given that the drivers of AMA departures are predominately systems based (eg, prolonged care delays), solutions focused on the clinical environment should be prioritized to reduce this outcome.

Registered nurses are the health care professionals most frequently attending to patients in hospitals, ensuring that they receive timely care interventions, surveillance, and communication about care plans.[7] Previous evidence demonstrates that favorable hospital work environments, characterized by nurse staffing adequacy, nurse-physician collegial relations, nurse autonomy, supportive nurse management, and a foundation for quality care, are associated with more timely ED care as measured by shorter patient lengths of stay and lower left without being seen rates.[8,9] Evidence also demonstrates that when nurses care for fewer patients at a time, patients are less likely to be readmitted or die in the hospital and more likely to have shorter lengths of stay.[10-12]

As emergency departments experience increased crowding,[13,14] patients are at a higher risk of care delays and suboptimal patient outcomes, including AMA departures. Nurse staffing is a modifiable factor in hospitals that may be leveraged to reduce rates of self-directed AMA departures. Patient experience and satisfaction are known drivers of AMA departures,[15] and favorable emergency nurse staffing may improve patient experience through nurses' increased availability to communicate and deliver care interventions. Evidence on reductions in patient AMA departures through improved nurse staffing, both in the emergency department and inpatient settings, has yet to be established.

This study aimed to evaluate the relationship between ED and adult medical and surgical inpatient patient-to-nurse staffing ratios and ED AMA departures. The rationale for examining both ED and medical-surgical staffing on AMA departures is that ED patient outcomes are influenced by processes in the inpatient setting, according to Asplin et al's[16] conceptual model. We hypothesized that when ED and inpatient medical and surgical nurses care for more patients at a time, hospital emergency departments experience higher AMA rates because delays in receiving care result from both waiting for an inpatient bed and delays in ED care.

Methods

In this cross-sectional study, 3 data sets were linked and merged using a common hospital identifier: the 2023 Penn Nurses4All survey, the Agency for Healthcare Research and Quality (AHRQ) Healthcare Cost and Utilization Project (HCUP) 2023 State Emergency Department Database (SEDD) from New Jersey (NJ) and Oregon (OR), and the American Hospital Association (AHA) Annual Hospital Survey.

The Nurses4All survey provided information on patient-to-nurse staffing ratios and nurse demographics including position and hospital unit. The AHRQ HCUP 2023 SEDD data set is an all-payer patient claims data set that provides information on ED patient encounters, patient demographics, primary diagnosis in the emergency department, and disposition used to determine whether an ED patient left AMA or not. The AHA Annual Hospital Survey provided hospital characteristics and the hospital identifier to facilitate data set linkages. This study was approved by the University of Pennsylvania Institutional Review Board.

CONCEPTUAL FRAMEWORK

Asplin's input-throughput-output model[16] informed the study objectives, outcome and predictor variables, and statistical analyses. This model states that patient outcomes and operational failures such as ED overcrowding are not solely driven by factors in the ED work environment, but also the inpatient environment. Multiple inpatient hospital factors influence ED outcomes as posited in this model,

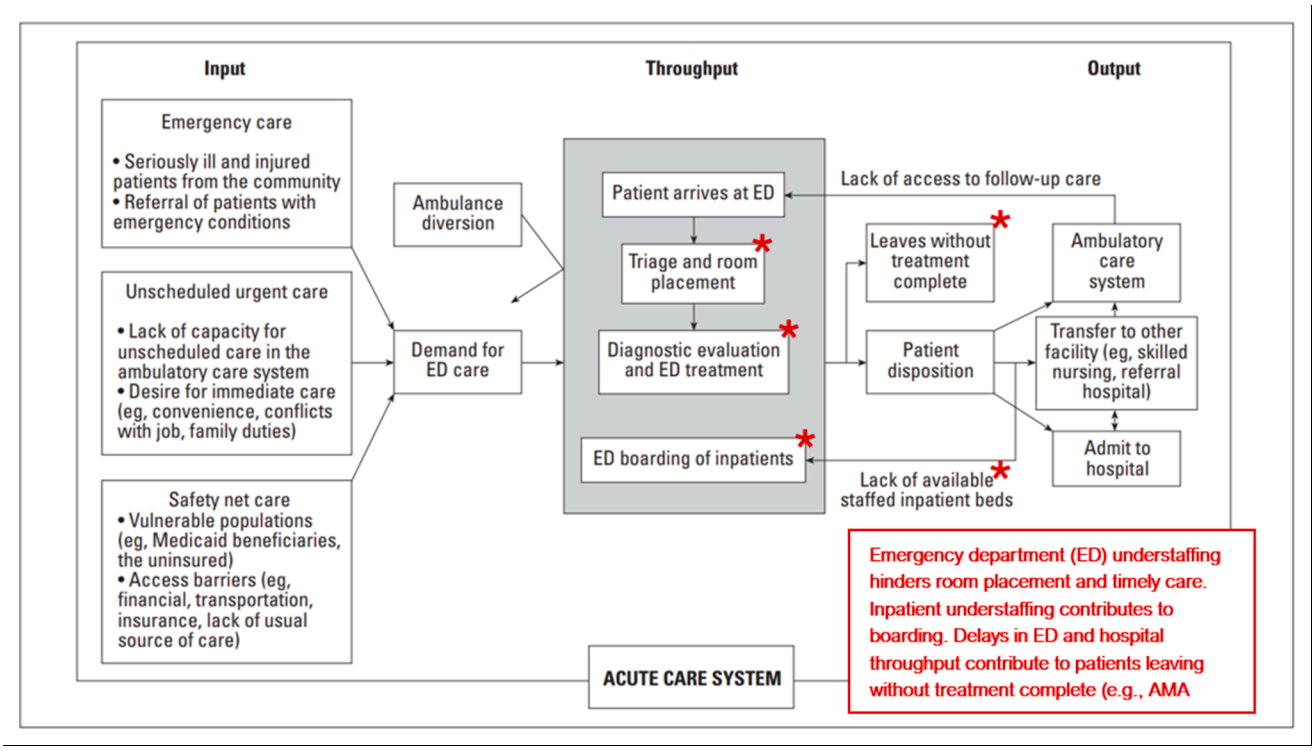

FIGURE 1

Nurse staffing impact in the input-throughput-output conceptual model. (Reprinted from "A conceptual model of emergency department crowding," Vol 42, Issue 2, Annals of Emergency Medicine, 2003, Asplin et al,[16] with permission from Elsevier.) AMA, against medical advice.

including inpatient discharges, bed availability, and workforce staffing. Previous evidence demonstrates that inpatient nurse work environments are predictive of emergency nurse job outcomes and patient care quality and safety,[17] as well as ED left without being seen rates and length of stay.[8] Given these interrelated factors, we examined the association of ED and medical-surgical nurse staffing on ED AMA rates in the present study. The impact of ED and medical-surgical (inpatient) nurse staffing in Asplin's conceptual model is presented in Figure 1.

SURVEY APPROACH

The 2023 Penn Nurses4All survey is a multistate electronic survey of registered nurses licensed to work in 9 states (California, Florida, Illinois, Louisiana, New Jersey, New Mexico, New York, Oregon, Washington), in addition to nurses in Pennsylvania recruited via mail and selected via a random sample. The survey was conducted between December 2023 and March 2024. The survey yielded responses from nurses across various health care settings (eg, hospitals, long-term care, schools); however, hospital nurses were the focus of the present study. The methodology of the survey is detailed elsewhere.[18,19]

Pertinent to the current research objective were staff nurses in emergency departments and on medical-surgical units who responded to the survey question: *how many patients were assigned to you on your last shift?*

SAMPLE

The primary sample in the 2 states included 336 emergency nurses in 71 hospitals, and the secondary sample was 775 medical-surgical nurses only in 74 hospitals. The patient sample included patients aged 18 to 90 years presenting to an emergency department in New Jersey or Oregon in 2023. We excluded patients who died during an ED visit and those whose ED diagnosis was classified as a "mental, behavioral, and neurodevelopmental disorder."[20] Patients presenting with behavioral health concerns were excluded, given that their reasons for departing AMA (or not) may be driven more by their clinical presentation and legal factors

TABLE 1
Hospital characteristics (n = 74 hospitals)

Hospital characteristic	n	%/SD
Teaching status		
Nonteaching	28	38
Minor	31	42
Major	15	20
Technology status		
Not high	47	64
High	27	36
Bed size		
≤250	39	53
251-500	21	28
>500	14	19
Annual ED volume, mean (SD)	50,357	31,318
State		
NJ	34	48
OR	37	52
Emergency nurse staffing ratio, mean (SD)	6.8	1.9
Medical-surgical nurse staffing ratio, mean (SD)	6.3	1.0

NJ, New Jersey; OR, Oregon.

TABLE 2
Patient encounters characteristics (n = 2,211,244 ED visits)

Patient characteristic	n	%/SD
Age, mean (SD)	50.1	19.5
Sex		
Male	954,725	43.2
Female	1,256,519	56.8
Race		
Asian or Pacific Islander	67,785	3.1
Black	363,758	16.5
Hispanic	426,121	19.3
Native American	14,682	0.7
Other	94,750	4.3
White	1,244,148	56.3
Insurance type		
Medicare	565,580	25.6
Medicaid	592,904	26.8
Private insurance	720,612	32.6
Self-pay or hospital responsibility	148,822	6.7
No charge	43,490	1.9
Other	139,836	6.3
Income level		
$1-$57,488	493,407	22.3
$57,489-$72,792	434,988	19.7
$72,793-$95,333	625,181	28.3
$95,334+	657,668	29.7
Arrival on weekend	593,462	26.8
Departure against medical advice	54,955	3.0
Patient comorbidities, mean (SD)	1.0	0.9

Percentages may not equate to 100 owing to rounding.

related to ED care. The HCUP SEDD data do not provide individual patient identifiers across both states, nor dates for the ED encounters; therefore, we were unable to identify circumstances where a unique patient had multiple ED visits.

VARIABLES

Primary Independent Variable

The independent variable of interest was nurse staffing, defined as the patient-to-nurse staffing ratios in emergency departments and, separately, medical-surgical units. Our primary focus was on emergency nurses, with a secondary analysis of medical-surgical nurses. In the Nurses4All survey, respondents were asked, "How many patients were assigned to you on your last shift?" The measure demonstrated moderate reliability based on an intraclass correlation coefficient (intraclass correlation coefficient [2] = 0.50) when including at least 3 nurses per hospital. Nurses provided a whole number that was then aggregated to the hospital level. In particular, individual ED and (separately) medical-surgical nurse staffing ratios were averaged to generate the 2 hospital-level staffing measures.

Outcome Variable

The primary outcome in this study was a binary variable indicating whether a patient departed AMA during an ED encounter, derived from the AHRQ HCUP SEDD data on patient disposition status.

Covariates

Patient covariates used in risk adjustment included the following variables typically evaluated in ED health services research and pertinent to patient acuity,[21,22] derived from the AHRQ HCUP SEDD data files: demographics (eg, age, race, sex); insurance payer; ED International

TABLE 3
Multivariate logistic regression modeling examining the impact of nurse staffing ratios on emergency department self-directed departures against medical advice

Variable	Model 1, unadjusted			Model 2, adjusted for patient and hospital characteristics		
ED staffing	**OR**	**95% CI**	***P* value**	**OR**	**95% CI**	***P* value**
Left AMA	1.21	1.02-1.44	<.05	1.19	1.09-1.31	<.001
Medical-surgical staffing	**OR**	**95% CI**	***P* value**	**OR**	**95% CI**	***P* value**
Left AMA	1.41	1.06-1.88	<.05	1.55	1.33-1.81	<.001

AMA, against medical advice; OR, odds ratio.
The independent variable is the patient-to-nurse staffing ratio. The outcome of departures against medical advice is interpreted as an OR. Models are adjusted for patient age, sex, race, insurance, chief complaint, number of comorbidities, weekend arrival (yes/no), annual income, hospital teaching status, technology capabilities, bed size, annual patient ED volume, and Elixhauser comorbidities. The emergency nurse staffing ratio includes 2,146,174 patients in 71 hospitals. The medical-surgical nurse staffing ratio includes 2,211,244 patients in 74 hospitals.

Classification of Diseases, 10th Revision diagnosis code; comorbidity count; Elixhauser comorbidities; receipt of care on a weekend (yes or no); and income level. Patient International Classification of Diseases, 10th Revision, ED diagnosis code was categorized into 1 of 20 categories using the AHRQ Clinical Classification Software.

Hospital covariates included teaching status (nonteaching, no medical fellows/residents; minor teaching, <1 fellow/resident per bed; major teaching, ≥1 fellow/resident to 4 beds), technology capabilities (high or not) indicating the capacity to provide transplant services or open heart surgery, annual ED patient volume, and bed size (≤250, 251-500, >500), all derived from the AHA Annual Survey.

STATISTICAL ANALYSES

Descriptive statistics including means, percentages, and SDs were computed for patient characteristics (demographics, percentage of ED encounters resulting in AMA departure, comorbidities), patient-to-nurse staffing ratios, and hospital characteristics. We include hospital characteristics for the maximum sample size included in the study, which were the models with medical-surgical nurses (n = 74; see Table 1). All variables had <15% missing, and missing variables for model covariates were dropped. Normality for continuous variables was assessed using data visualization techniques including histograms. Assumptions for logistic regression modeling were evaluated (eg, use of a binary outcome variable, no multicollinearity of predictors).

Logistic regression models with robust SEs to account for the clustering of patients and nurses within hospitals were fitted to estimate the odds of a patient AMA departure associated with 1 additional patient in the nurse staffing ratio. Two separate models were fitted: 1 for ED and 1 for medical-surgical nurse staffing. SEs were computed using the Huber-White[23] procedure to correctly account for the clustering of patients and nurses within hospitals. These models were adjusted for patient and hospital covariates outlined in the Methods. There were 71 hospital clusters in the models assessing ED staffing and 74 in the models assessing medical-surgical staffing.

Sensitivity analyses are provided in the Supplementary File, including an analysis of patients with behavioral health conditions and models that control for state fixed effects. We also conducted a mixed-effects logit model using data from NJ patients only where patient identifiers were available to account for potential multiple visits by patients within the same hospital.

Results

The final sample included 2,211,244 visits for patients seen and discharged from an emergency department in 74 New Jersey and Oregon hospitals (Table 2). On average, patients were 50 years of age (SD = 20) and more likely to be female (57%) and white (56%) with Medicaid (27%) or Medicare insurance (26%). On average, 54,955 visits resulted in a departure AMA (3%). Patients most frequently were in the highest income quartile with an annual income of at least $72,793. The category of patients' ED primary diagnosis, as well as the Elixhauser comorbidities included for risk adjustment in the models, is presented in Supplementary Table 1.

Hospitals (Table 1) were most frequently nonteaching (38%) or minor teaching hospitals (42%), which were not

high technology (64%) with ≤250 beds (53%). The mean patient-to-nurse staffing ratio for emergency nurses was 6.8 (SD = 2) and 6.3 (SD = 1) for medical-surgical nurses.

The logistic regression models demonstrating that patient-to-nurse staffing ratios predicted self-directed departures AMA are presented in Table 3. These models can be interpreted as meaning, for example, that each additional patient per emergency nurse was associated with a 21% increase in the odds of an AMA patient departure in the unadjusted model for emergency nurse staffing ratios (model 1, OR, 1.21; 95% CI, 1.02-1.44; $P < .05$). These findings persisted after adjusting for patient and hospital characteristics (model 2, OR, 1.19; 95% CI, 1.09-1.31; $P < .001$).

The medical-surgical patient-to-nurse staffing models (Table 3) demonstrated similar results. Each additional patient per medical-surgical nurse was associated with a 41% increase in the odds of an AMA patient departure in the unadjusted models (OR, 1.41; 95% CI, 1.06-1.88; $P < .05$) and a 55% increase in the odds of an AMA departure after adjusting for hospital and patient characteristics (OR, 1.55; 95% CI, 1.33-1.81; $P < .001$).

In Supplementary Tables 2 to 5, our sensitivity analyses uphold the study findings demonstrating an increase in patient AMA departures for each additional patient added to a nurse's workload.

Discussion

A patient departure AMA from the emergency department is an adverse event for patients and hospitals, given that patients may be terminating or deferring care services that are necessary. This outcome threatens hospitals' primary mission to deliver quality health care services to patients in the community. Patients who leave ED visits AMA are more likely to be sicker and return to hospitals for repeat ED or inpatient care than patients who do not leave AMA during their stay.[1,3] Overall, our findings identify a systems-based, modifiable factor (nurse staffing) that can be leveraged by hospitals to reduce a costly patient outcome, which may, in some cases, be avoidable. These findings contribute new knowledge to the literature base in that each additional patient added to a nurse's workload in both emergency departments and medical-surgical units was associated with higher odds of AMA departures. In other words, the adequacy of nurse staffing in both the emergency department and inpatient units influences the discharge outcomes of ED patients.

Our findings confirm similar studies demonstrating an association among the hospital nurse work environment, patient left without being seen rates, and length of stay.[8,9] In more favorable work environments where employers invest in lower staffing ratios, interprofessional collaboration, leadership, and quality care, hospitals experience lower patient left without being seen rates.[8] It is preferable for hospitals to retain patients whom clinicians have identified as needing additional care to stay for the duration of their treatment. More than half of the patients in the study were insured by Medicare or Medicaid, which could pose a challenge for hospitals' reimbursement payments given that patients who leave AMA are more likely to incur more hospital readmissions than those who do not leave AMA.[5] Patients are safer and receive higher quality care when their visit to an emergency department for care, treatment, and surveillance is completed and they are less likely to return for a repeat ED or hospital stay.[3,6,15]

Limitations

Given that the study design is cross-sectional, causal linkages cannot be inferred from the study findings. Our data were collected before Oregon passed a minimum nurse staffing requirement policy and thus identified an opportunity for a pre- and postimplementation investigation. The AHRQ HCUP SEDD data were limited to patient visits (encounters) without patient identifiers; thus, we were unable to link AMA departures to patients' index ED visits. This is commonly reported in the health services literature[24,25] when using AHRQ HCUP data. The data do not facilitate linkages between the specific dates that nurses took the survey and the dates that patients received care in the emergency department. Thus, our study does not explore at what point a patient decided to leave AMA during the visit given limitations with available variables. The study outcome variable was derived from the AHRQ HCUP data and includes patients who "Left Against Medical Advice or Discontinued Care"; thus, it is unknown whether patients signed an official AMA departure form during their visit. Patients in the study could present multiple times during the study year. Although we were unable to determine the dates when patients presented for care, a mixed-effects logit model (Supplementary File) upheld similar results to our main study findings, suggesting potentially minimal concerns for multicollinearity.

Implications for Emergency Nursing

The results from our study indicate that nursing care in both the emergency department and inpatient settings predicted ED patient AMA departures. Furthermore, our findings

support and further validate Asplin's input-throughput-output conceptual model, particularly supporting the relationships between nurse staffing, unavailability of inpatient beds, and ED boarding that Asplin et al[16] propose. When inpatient nurses care for too many patients at 1 time, their capacity to accept new admissions or transfers from the emergency department in a timely manner becomes extremely limited, resulting in a backlog and ED boarding. The backlog of the emergency department contributes to higher rates of patients leaving AMA and leaving without being seen.[13,26] It is less surprising that the nurse staffing ratios of emergency nurses also contribute to higher rates of AMA: when emergency nurse workloads are high, they cannot fully attend to all patients on their unit.

Overall, our results emphasize that nurse workloads may be associated with potentially avoidable and adverse patient AMA departures. Nurse staffing ratios are a modifiable feature of hospitals, and states are increasingly enacting policies around them given the substantial evidence base demonstrating better outcomes associated with safer nurse workloads.[9,27] California and Oregon[28,29] are 2 states that have successfully passed and implemented legislation mandating minimum nurse staffing requirements in hospitals, wherein hospitals must staff enough nurses to achieve a ratio of 5 nurses per patient on medical-surgical units, for example.

California's nurse staffing law has contributed to improved patient and nurse outcomes sustained over time. For example, patients in California receive on average 3 additional nursing care hours per day in comparison with patients in other states.[28] Nurses in California experience lower odds of burnout than those practicing outside of California, attributed in part to the improved staffing ratios in the state.[30] Oregon only recently passed nurse staffing legislation after this study data were collected, thus presenting an opportunity for further longitudinal policy evaluation.

Improving hospital nurse staffing is not the full solution to the multifaceted problem of patients leaving AMA. However, it could make a significant difference as suggested by our study results and is an underappreciated approach within the control of hospitals. ED administrators are often opposed to mandated hospital nurse staffing policies for fear of exacerbating crowding that manifests largely in the emergency department. Our results suggest that ED clinicians should be advocates for minimum safe nurse staffing policies given the evidence in this paper that adequate nursing resources may facilitate better patient throughput.

Most evidence demonstrating a relationship between high patient-to-nurse staffing ratios and suboptimal patient outcomes is established in the inpatient setting.[31,32] Using the evidence from this study, stretcher-side emergency nurses can be advocates for safer patient-to-nurse staffing ratios at their local hospitals and at the state and federal levels given this evidence. For example, advocating for the passage of the Addressing Boarding and Crowding in the Emergency Department Act (HR 2936)[33] using evidence from this study is a potential strategy for nurses to advance safer care environments for clinicians and patients. Future research is needed to explore the specific reasons behind patient AMA departures and the timing during their stay (eg, during the triage process or while a patient has been roomed).

Conclusion

Higher patient-to-nurse staffing ratios in emergency departments and medical-surgical units are associated with increased patient AMA departures. Our findings support that ED patient outcomes are not solely influenced by the nursing resources in the emergency department, but also the inpatient setting. Nurse staffing is a modifiable feature of hospitals that employers must invest in systematically across hospital units to reduce adverse and costly patient outcomes, such as AMA departures.

Author Disclosures

Conflicts of interest: none to report.

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Supplementary Data

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SUPPLEMENTARY TABLE 1
Emergency department diagnoses and elixhauser comorbidities (n = 2,211,244 ED visits)

Clinical category of ED diagnosis	N	(%)
Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism	14,430	0.7
Diseases of the circulatory system	219,422	10
Diseases of the digestive system	169,950	8
Diseases of the ear and mastoid process	22,082	1
Endocrine, nutritional, and metabolic diseases	39,182	2
Diseases of the eye and adnexa	22,123	1
Factors influencing health status and contact with health services	48,642	2
Diseases of the genitourinary system	161,851	7
Certain infectious and parasitic diseases	60,460	3
Injury, poisoning and certain other consequences of external causes	427,721	19
Congenital malformations, deformations and chromosomal abnormalities	467	0.02
Diseases of the musculoskeletal system and connective tissue	211,222	10
Neoplasms	8,565	0.4
Diseases of the nervous system	108,468	5
Pregnancy, childbirth, and the puerperium	72,823	3
Diseases of the respiratory system	127,846	6
Diseases of the skin and subcutaneous tissue	80,212	4
Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified	415,778	19
Elixhauser comorbidities		
Acquired immune deficiency syndrome	9,020	0.4
Alcohol abuse	25,715	1
Autoimmune conditions	20,733	1
Leukemia	2,142	0.1
Lymphoma	2,752	0.1
Metastatic cancer	6,375	0.3
Solid tumor, in situ	120	0.01

continued

SUPPLEMENTARY TABLE 1
Continued

Clinical category of ED diagnosis	N	(%)
Solid tumor, malignant	13,519	0.6
Dementia	31,716	1.4
Depression	65,828	3.0
Diabetes with chronic complications	74,225	3.4
Diabetes without chronic complications	167,126	7.6
Drug abuse	23,643	1.1
Hypertension, complicated	100,722	4.6
Hypertension, uncomplicated	427,755	19.3
Chronic pulmonary disease	161,051	7.3
Obesity	70,027	3.2
Peripheral vascular disease	19,190	0.8
Hypothyroidism	64,895	3.1
Other thyroid disorders	11,142	0.5

Note. ED=emergency department. ICD-10 diagnoses were aggregated using the AHRQ Clinical Classifications Software. Select Elixhauser comorbidities and AHRQ Clinical classification disease (eg, Diseases of the perinatal period) were omitted from the adjusted study models due to small cell sizes. Percentages may not equal 100 due to rounding.
Sensitivity Analysis.

SUPPLEMENTARY TABLE 2

Odds Ratios of ED Patient Departures Against Medical Advice and Nurse Staffing Ratios Including Patients with Behavioral Health Conditions

	Model 1 Unadjusted			Model 2 Adjusted for patient and hospital characteristics		
ED staffing	**OR**	**95% CI**	**P-value**	**OR**	**95% CI**	**P-value**
Left AMA	1.20	1.01-1.43	<.05	1.18	1.09-1.30	<.001
Med-surg staffing	**OR**	**95% CI**	**P-value**	**OR**	**95% CI**	**P-value**
Left AMA	1.39	1.06-1.85	<.05	1.52	1.31-1.78	<.001

Note. AMA=departure against medical advice; ED=emergency department; OR=odds ratio. Independent variable is patient-to-nurse staffing ratio. Outcome of departures against medical advice are interpreted as odds ratios. Models are adjusted for patient age, sex, race, insurance, chief complaint, number of comorbidities, weekend arrival (yes/no), annual income, hospital teaching status, technology capabilities, bed size, annual patient ED volume, and Elixhauser comorbidities.

SUPPLEMENTARY TABLE 3

Odds Ratios of ED Patient Departures Against Medical Advice and Nurse Staffing Ratios Adjusted for State Effects

	Model 1 Unadjusted			Model 2 Adjusted for patient and hospital characteristics		
ED staffing	**OR**	**95% CI**	**P-value**	**OR**	**95% CI**	**P-value**
Left AMA	1.21	1.02 - 1.44	<.05	1.12	1.00-1.26	<.05
Med-surg staffing	**OR**	**95% CI**	**P-value**	**OR**	**95% CI**	**P-value**
Left AMA	1.41	1.06 - 1.88	<.05	1.48	1.21-1.81	<.001

Note. AMA=departure against medical advice; ED=emergency department; OR=odds ratio. Independent variable is patient-to-nurse staffing ratio. Outcome of departures against medical advice are interpreted as odds ratios. Models are adjusted for patient age, sex, race, insurance, chief complaint, number of comorbidities, weekend arrival (yes/no), annual income, hospital teaching status, technology capabilities, bed size, annual patient ED volume, and Elixhauser comorbidities.

SUPPLEMENTARY TABLE 4

Odds Ratios of ED Patient Departures Against Medical Advice and Nurse Staffing Ratios: Independent contributions of ED staffing and Medical-Surgical Staffing

	Model 1 Unadjusted			Model 2 Adjusted for patient and hospital characteristics		
Staffing	**OR**	**95% CI**	**P-value**	**OR**	**95% CI**	**P-value**
ED	1.13	0.89-1.41	0.31	1.09	1.00-1.20	<.05
Med-Surg	1.27	0.87-1.85	0.21	1.47	1.25-1.73	<.001

Note. AMA=departure against medical advice; ED=emergency department; OR=odds ratio. Independent variable is patient-to-nurse staffing ratio. Outcome of departures against medical advice are interpreted as odds ratios. Models are adjusted for patient age, sex, race, insurance, chief complaint, number of comorbidities, weekend arrival (yes/no), annual income, hospital teaching status, technology capabilities, bed size, annual patient ED volume, and Elixhauser comorbidities. VIF for nurse staffing variables was <2.

SUPPLEMENTARY TABLE 5

Mixed-effects Logit Models Examining the Impact of Nurse Staffing Ratios on Emergency Department Self-Directed Departures Against Medical Advice

	Model 1 Unadjusted			Model 2 Adjusted for patient and hospital characteristics		
Staffing	**OR**	**95% CI**	**P-value**	**OR**	**95% CI**	**P-value**
ED	1.13	1.13-1.14	<.001	1.15	1.14-1.16	<.001
Med-Surg	1.32	1.31-1.33	<.001	1.32	1.31-1.33	<.001

Note. Outcome of departures against medical advice are interpreted as odds ratios. Models are adjusted for patient age, sex, race, insurance, chief complaint, number of comorbidities, annual income, hospital teaching status, technology capabilities, bed size, annual patient ED volume, and comorbidity count and are only conducted for the state of New Jersey due to the availability of an MRN variable in this state. VIF for nurse staffing variables was <2.